LUTHER JOHN ALLEN CONCEPCION

JUNIOR FULL STACK WEB DEVELOPER

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SUMMARY

Computer Science student and freelance web developer since 2025, with practical experience delivering client applications, full-stack web projects, and offline AI systems. Seeking junior full-stack web developer roles and internships in the Philippines or remotely. Expected graduation: August 2027.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL
Frontend: Next.js, React, HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, Zustand
Backend: Node.js, Fastify, FastAPI, Express.js, REST APIs, WebSockets
Databases: MySQL, PostgreSQL, SQLite, Supabase, Prisma, Redis
Tools & DevOps: Git, GitHub, Docker, Apache, Nginx, Vercel, Railway, Composer
AI & Data: OpenAI, local LLMs, RAG, FAISS, BM25, embeddings, cross-encoder reranking, OCR
Core Concepts: Database design, data processing, authentication, WebRTC, STUN/TURN, file encryption

EXPERIENCE

Freelance Web Developer **2025 - Present**
Independent | *Private Client*

- Independently delivered a Beneficiary Duplicate Detection System for a private client, completing development over three months and deploying it locally in March 2026.
- Handled frontend implementation, PHP backend development, database schema design, duplicate-matching logic, and offline deployment.
- Implemented workflows to process beneficiary records into the database and export records to Excel, supporting storage and duplicate checks for more than 10,000 records.
- Built record-review interfaces, analytics charts, and geographic visualizations using Bootstrap, jQuery, DataTables, Chart.js, and Leaflet.

PROJECTS

Beneficiary Duplicate Detection System **Deployed March 2026**
Solo Freelance Developer | *Private Client*

Stack: PHP 8.1+, MySQL, Apache, Bootstrap, jQuery, DataTables, Chart.js, Leaflet, PhpSpreadsheet, Composer

- Built exact matching and fuzzy string comparison to identify duplicate beneficiary records, including entries with typographical differences.
- Designed the database schema and implemented database access with PDO MySQL for record storage, review, and duplicate verification.
- Automated record processing into the database and Excel exports using PhpSpreadsheet; stored and checked more than 10,000 beneficiary records.
- Implemented data tables for reviewing records, a Chart.js analytics dashboard with data labels, and Leaflet geographic distribution maps with heatmap support.
- Deployed the application for local, offline use. The server environment supports PHP 8.1 or later, MySQL 5.7 or later or MariaDB 10.4 or later, and Apache.

Wild Dex **May 2026**
Solo Developer *wilddex-xi.vercel.app*

Stack: Next.js, React, TypeScript/TSX, JavaScript, Tailwind CSS, OpenAI, Google Authentication, Prisma, Supabase, Vercel

- Independently built and deployed a wildlife entertainment application with a Next.js interface styled using Tailwind CSS.
- Integrated camera-based wildlife identification using an OpenAI model to process camera photos.
- Implemented Google authentication, backend APIs, and data models for wildlife sightings and user activity.
- Used Prisma and Supabase for application data access and storage; developed features supporting sightings and competitive gamification.
- Deployed the application on Vercel for public access.

Peer Drop **June 2026**
Solo Developer *peer-drop.vercel.app*

Stack: Next.js 15, React, TypeScript, Tailwind CSS, Zustand, Node.js, Fastify, WebSockets, Redis, PostgreSQL, WebRTC, Coturn

- Independently built a browser file-sharing application with a Next.js frontend and a Node.js/Fastify signaling backend.
- Implemented connection through shareable session links, six-character room codes, and QR codes for mobile users.
- Added password-protected group rooms, host approval, and a waiting room to control participant access.
- Used WebRTC DataChannels for direct browser-to-browser transfers, with Coturn STUN/TURN support for NAT traversal and encrypted relay fallback.
- Implemented ephemeral ECDH key exchange and AES-256-GCM encryption for streamed file chunks.
- Used WebSockets for signaling, Redis for ephemeral sessions, and PostgreSQL for metadata; kept file contents out of server-side storage.
- Deployed the public frontend on Vercel.

The Archive - Thesis **June 2026**
Lead Developer

Stack: JavaScript ES6+, HTML5, CSS3, Python 3.10+, FastAPI, Uvicorn, PyMuPDF, Tesseract OCR, llama-cpp-python, Ollama, sentence-transformers, FAISS, BM25Okapi

- Developed an offline academic PDF question-answering system with clickable document/page citations, source highlights in an interactive PDF inspector, selectable reranking depth, and formatted four-sheet Excel audit reports.
- Built PDF parsing and OCR workflows using PyMuPDF and Tesseract, with local text embeddings and retrieval indexes persisted as .npy, .index, and .pkl files.
- Integrated Qwen 2.5 3B Instruct using GGUF Q4_K_M through llama-cpp-python on CPU SIMD/AVX2, with an Ollama phi3.5 fallback for local inference.
- Used sentence-transformers/all-MiniLM-L6-v2 for 384-dimensional, L2-normalized embeddings and FAISS IndexFlatIP for vector retrieval.
- Combined BM25Okapi and FAISS retrieval in parallel, merging results with Reciprocal Rank Fusion (k = 60) to select 30 candidate chunks.
- Applied a bi-encoder pre-filter to narrow results to 15 candidates, followed by ms-marco-MiniLM-L-2-v2 or L-6-v2 cross-encoder reranking and an adaptive score-gap cutoff; supplied the top three to five chunks to the local LLM.
- Recorded 90.0% Recall@5 against an 86.0% baseline. Recall on empirical-fact queries increased from 77% to 100%, an improvement of 23 percentage points.
- Recorded 93.2% RAGAS faithfulness and a 24.6% reduction in hallucinations; dense-literature evaluation reached statistical significance ($p = 0.0412$).

- Measured hybrid retrieval at approximately 48-73 ms, reranking at approximately 289 ms, and total CPU retrieval at approximately 610 ms, with memory usage below 8.5 GB.

EDUCATION

Bachelor of Science in Computer Science **2023 - Present**
Isabela State University Cauayan City Campus *Cauayan City, Isabela*

Expected graduation: August 2027

LEADERSHIP

Programming Club **2025 - Present**
Secretary *Isabela State University Cauayan City Campus*

Programming Club **2024 - 2025**
Business Manager *Isabela State University Cauayan City Campus*

CERTIFICATIONS

WordPress Website Development Course **July 2026**
LearnKartS via Coursera

Building Web Applications in PHP **November 2025**
University of Michigan via Coursera